

Third Semester

Advanced Java (MCC- 2.3.1)

Full marks–100(Mid Semester: 20 Marks; End Semester: 80Marks)

Objectives:

The objective of the course is to give an idea on advance java and its implementation in the programming perspective.

At the end of the course, the students will be able to:

- Understand basic knowledge advance java, web programming and concepts of UI programming.
- Database connectivity, JSP, MVC architecture.
- Servlet and RMI application and database program.
- Java bean, EJB, Ajax
- Spring and hibernate framework.

Course Outcomes:

Upon successful completion of this course, the student shall be able to do programming and Implementation on:

- Advance java, web programming and concepts of UI programming.
- Database connectivity, JSP, MVC architecture.
- Servlet and RMI application and database program.
- Java bean, EJB, Ajax.
- Spring and hibernate framework.

Unit I

Basic concept of network, types and features, browser, client, server, URL, webpage and website, Web Programming, HTML, basic structure, basic HTML commands, list, table, image, hyperlink, heading, marquee, form and components, frame, audio and video in HTML page, DHTML, program structure, java script, syntax structure, datatypes, tokens, operator, control structure, array, function, event and types, class and exception, process of writing and running program (HTML and DHTML), CSS, properties, CSS syntax, CSS selector, types of CSS, basics of XHTML and XML.

Unit II

JDBC, program on JDBC (insert, update, delete, select), JDBC use on application program, applet, AWT, swing (MVC), Enterprise Java Programming, JSP, features, syntax structure, lifecycle, dynamic webpage creation by jsp, Anatomy of JSP Page, Scripting Elements, implicit objects, directive elements, MVC in JSP, Session Tracking, jsp database access, JSP exceptions

Unit III

SERVLET, features, jsp vs. servlet, servlet syntax structure, servlet life cycle, types of servlets, dynamic page creation by servlet, database access using servlet, session tracking, servlet API, servlet packages, creating cookies, RMI, features, RMI structure, Marshaling and Unmarshaling, RMI application and database program.

Unit IV

JAVA BEAN, features, properties, types of java bean, program how to create java bean, EJB, features of EJB, life cycle of EJB, types of EJB, EJB architecture, program for creating EJB, AJAX, features, working of AJAX, AJAX request and response example, AJAX database connectivity. Spring, spring framework, features, spring modules, spring example program, steps for creating spring application, Struts, features, creating struts application, basic features of JSF, FACELET, JSTL, JMS, hibernate framework.

Books:

Reference Books:

1. Ivan Bayross, part-1, web technologies, bpb publication, New Delhi
2. Ivan Bayross, part-2, web technologies, bpb publication, newdelhi
3. Steven Holzner, Java8ProgrammingBlackBook, Dreamtech Press
4. Eric Jendrock, Java EE6 basic concept, Pearson
5. Java(j2EE) and Spring Framework, Ayan International
6. HTML&CSS: The Complete Reference, Fifth Edition. Thomas A. Powell

Machine Learning (MCC-2.3.2)

Full marks–100 (Mid Semester: 20Marks; End Semester: 80Marks)

Objectives:

- Enable students to realize the role of statistical frameworks for analysis, design and visualization of data.
- To make the students technically aware of state-of-the-art in the field of ML.
- To enable students to understand various intrinsic behavior of data and the logical framework behind the learning models.
- To understand the basic theory underlying machine learning.
- To be able to apply machine learning algorithms to solve problems of moderate complexity.
- To apply the algorithms to a real-world problem, optimize the models learned and report on the expected accuracy that can be achieved by applying the models.
- To understand a range of machine learning algorithms along with their strengths and weaknesses.

Course Outcomes:

- The course provides handy approach for fundamental model building and prediction.
- Computation of performance metrics of the developed models using different datasets.
- To enable students for incorporating their own customized learning frameworks from the above knowledge.
- Develop an appreciation for what is involved in Learning models from data
- Apply the algorithms to a real problem, optimize the models learned and report on the expected accuracy that can be achieved by applying the models
- Understand how to evaluate models generated from data
- Understand a wide variety of learning algorithms

Unit I: Supervised Learning 1:

Overview of Supervised Learning, Classification and Regression problems, Nearest Neighborhood (KNN) Classifier, Naïve Bayes Classifier, Multiple Linear Regression, Shrinkage Methods (Ridge Regression, Lasso Regression, Elastic Net), Logistic Regression, Linear Discriminant Analysis, Feature Subset Selection, Loss Function, Test and Training Error, Bias, Variance, Model Complexity, Bias-Variance Trade-off, Cross- Validation, Bootstrap Methods,

Unit II: Supervised Learning 2:

SVM for Binary and Multiclass Classification, SVM for Regression, Regression and Classification Trees, Ensemble Learning and Random Forest. Boosting Methods (AdaBoost, XG boost), Performance of Classification Algorithms (Confusion Matrix, Precision, Recall and ROC Curve).

Unit IV: Unsupervised Learning:

Clustering (K-means, spectral clustering), Feature Extraction (Principal Component Analysis (PCA), kernel PCA, Independent Component Analysis (ICA)), Non-negative Matrix Factorization and Collaborative Filtering, Mixture of Gaussians and its Applications to Clustering, Anomaly Detection, Novelty Detection, Learning Vector Quantization (LVQ)

Unit V: Neural Networks:

Models of a Neuron, Network Architectures, Least-Mean-Square algorithm, Perception, Perceptron Learning Algorithm, Multilayer Perceptron, Back-Propagation Algorithm (BPA), Heuristics for making the BPA perform better, accelerated convergence of BP Learning, Conjugate Gradient Method, Quasi-Newton Methods, Self-Organizing Maps, Associative Network: Autoassociative Memory, Heteroassociative Memory, Bidirectional Associative Network

Lab implementation of the above methods and algorithms using Python.**Books:****Texts:**

1. Trevor Hastie, Robert Tibshirani, Jerome Friedman , *The Elements of Statistical Learning-Data Mining, Inference and Prediction* ,Second Edition , Springer Verlag, 2009.
[chapters: 2,3(3.1-3.4,3.6),4(4.3-4.4),7(excluding 7.8),8.5,9(9.2),(10.1,10.4-10.5,10.8,10.10,10.14),13(13.1-13.3),14(14.3.1,14.3.6-14.3.8,14.5,14.6-14.8),15]
2. S. Haykin, *Neural Network- a comprehensive Foundation*, Pearson Education, Second Edition, 1999. Chapters 1(:1.1-1.3,1.6,)3(3.5,3.8,3.9)4(,4.1-4.5,4.6,4.17,4.18), 6,9.
3. Andreas C. Muller and Sarah Guido, *Introduction to Machine Learning with Python-A guide for data scientists*, O'Reilly, (SPD), 2017 (**For lab only**) [Relevant portions }

References

1. Gareth James. Daniela Witten. Trevor Hastie. Robert Tibshirani -*An introduction to statistical learning with applications in R*, Springer, 2013.
2. C. M. Bishop –*Pattern Recognition and Machine Learning*, Springer, 2006.
3. Ethem Alpaydin, *Introduction to Machine Learning*, PHI, 2010.
4. Tom M. Mitchell, *Machine Learning*, McGraw Hill Education, Indian Edition, 2013. Chapter 6(6.9)

Theory of Computation (MCC-2.3.3)

Full marks–100(Mid Semester: 20 Marks; End Semester: 80 Marks)

Objectives:

The primary objective of this course is to introduce the basic principles, techniques, and applications of automata theory concepts in real world problems.

- To understand the concept of machines: finite automata, pushdown automata, linear bounded automata, and Turing machines.
- To understand the formal languages and grammars: regular grammar and regular languages, context-free languages, and context-free grammar; and introduction to context-sensitive language and context-free grammar, and unrestricted grammar and languages.
- To understand the relation between these formal languages, grammars, and machines.
- To understand the complexity or difficulty level of problems when solved using these machines.
- To understand the concept of algorithm.
- To compare the complexity of problems.

Course Outcomes:

Upon successful completion of this course, the student shall be able to understand:

- able to design Finite Automata machines for given problems
- Describe the types of grammar and derivation tree
- able to analyze a given Finite Automata machine and find out its Language
- able to design Pushdown Automata machine for given CF language(s)
- able to generate the strings/sentences of a given context-free languages using its grammar · able to design Turing machines for given any computational problem
- Develop a computational model using Turing machine for the given problem. · Examine the complexity for P and NP completeness for the given problem.

Unit I

Introduction to finite automata, Central concepts of automata theory, Informal picture of finite automata, Deterministic finite automata, non-deterministic finite automata, Application, Formal Language.

Unit II

Regular expressions, Finite automata and Regular expressions, Applications of regular expressions, Algebraic laws of regular expressions, Pumping Lemma and its application for

regular languages, Closure and Decision properties of regular languages.

Unit III

Context-Free Grammars, Parse trees, Ambiguity in Grammar & Languages, Pushdown automation. The language of PDA. Equivalence of PDA's and CFG's. Deterministic pushdown automata, Chomsky Normal form, the pumping Lemma for context free languages, Decision properties of CFL's.

Unit IV

The Turing machine, Programming techniques for Turing machines, Extension to the basic Turing machine, Restricted Turing machine, Turing machines and computers.

Non-Recursively enumerable languages, Undecidable problem that in recursively enumerable, Undecidable problem about Turing machines, Post's correspondence problem, other undecidable problems.

Books:

Texts:

1. Introduction to Automata Theory, Languages and Computation- J.Hopcroft, R. Motwani, J. D. Ullman– Pearson Education

Reference Books:

1. Introduction to Theory of Computation– M.Sipser, Thomson Learning
2. P. Linz, “An Introduction to formal Languages and Automata”, Norasa, 2000
3. Lewish Papadimitra: Theory of Computations, Prentice Hall of India, New Delhi.

Elective-A07

Computer Graphics (MCC -2.3.4)

Full marks–100 (Mid Semester: 20Marks; End Semester: 80Marks)

Objectives:

The primary objective of computer graphics is to understand how computer creates and manipulates on image and how it displays objects graphically. It provides an opportunity to learn how to rescale, transmit, shear, and rotate different graphical objects.

The basic objective on the course is-

- To understand the concept of representation of information in graphical and pictorial way.
- Different types display of information by computer.
- Two dimension and three-dimension transformation and translation.

Course Outcomes:

Upon successful completion of this course, the student shall be able to understand:

- Introduction and applications of computer graphics
- Video display devices, Cathode Ray Tube implementation principles.
- Raster Scan Displays, Random Scan Displays operating principles.
- Line drawing Algorithms, Mid-point circle algorithm and Ellipse generating algorithms with examples.
- Transformation and Translation in 2d and 3d with problems.
- Rotation, Scaling and Reflection in 2d and 3d.
- Shearing and problem based on 2d and 3d transformation.
- Homogeneous co-ordinates and composite transformation. Viewing Pipeline and Viewing coordinate reference frame.
- Line Clipping and Polygon Clipping with problems. Polygon and Quadratic Surfaces.
- Spline Representation, Bezier Curves and Surfaces with derivation. B-Spline Curves and Surfaces with derivation.
- Three-dimensional Viewing Pipeline. Viewing Coordinates and Parallel and Perspective Projections.
- Classification of Visible-Surface Detection methods and Algorithms with example.
- Difference between image space and object space method, Back- Face Detection Algorithms with example.
- Depth-Buffer Method Algorithms with example, A-Buffer Method Algorithms with example

- Basic Illumination Models, Displaying Light Intensities. Halftone Patterns and Dithering Techniques.
- Gouraud Shading Polygon-Rendering and Phong Shading Polygon-Rendering Methods and derivation with example.

Unit I

A survey of computer graphics: Computer Aided Design, Presentation Graphics, Computer Art, Entertainment, Education Training, Visualization, Image Processing, Graphical User Interface. Overview of graphics system: Video Display Devices, Raster Scan Systems, Random Scan Systems, Input Devices, Hard-copy Devices, Graphics Software. Output primitives: Points and lines, DDA and Bresenham's Line Drawing Algorithm, Midpoint circle algorithm, Filled area primitives. Attributes of output primitives: Line attributes, Curve attributes, Color and grayscale levels, Area-fill attributes, Character attributes, Bundled attributes.

Unit II

Two dimensional geometric transformations: Basic Transformation (Translation, Rotation, Scaling), Matrix representation and homogenous coordination, Composite Transformation, Reflection, Shear. Two-dimensional viewing: The viewing Pipeline, Viewing Coordinate Reference frame, Window-to-view port coordinate transformation. Clipping: Line Clipping (Cohen Sutherland Algorithm), Polygon clipping (Sutherland- Hodge men Algorithm).

Unit III

Three-dimensional object representation: Polygon Surfaces, Quadratic surfaces, Spline Representations, Beizer Curves and surfaces, B-Spline Curves, and surfaces.

Three dimensional geometric and modeling transformations: Translation, Rotation, Scaling, Reflections, Shears, Composite Transformation. Three-dimensional viewing: Viewing pipeline, Viewing coordinates, Projections (Parallel and Perspective).

Unit IV

Visible- surface detection methods: Classification of visible-surface detection algorithms, back-face detection, Depth-Buffer method, A-Buffer method, Scan-line method, Depth-sorting method. Illumination Models: Basic illumination models, displaying light intensities, halftone Patterns and Dithering Technique, Polygon Rendering Methods, (Gouraud and phong shading)

Textbook: -

2. Computer Graphics C Version, by D. Heam and M. P. Baker, 2nd Edition, Pearson Education, 2002.

Elective -A08

Artificial Intelligence (MCC -2.3.4)

Full marks–100(Mid Semester: 20 Marks; End Semester: 80 Marks)

Objectives:

The primary objective of this course is to introduce the basic principles, techniques, and applications of Artificial Intelligence, emphasizing on the teaching of these fundamentals, instead of providing an expertise of specific software tools or programming environments. The focus on the course is-

- To gain a historical perspective of AI and its foundations.
- To become familiar with basic principles of AI toward problem solving, inference, perception, knowledge representation, and learning.
- To Investigate applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.
- To understand basics of machine learning model for simulation and analysis.
- To Explore the current scope, potential, limitations, and implications of intelligent systems.

Course Outcomes:

Upon successful completion of this course, the student shall be able to:

- Demonstrate fundamental understanding of the history of artificial intelligence (AI) and its foundations.
- Apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.
- Demonstrate awareness and a fundamental understanding of various applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.
- Demonstrate proficiency developing applications in an 'AI language', expert system shell, or data mining tool.
- Demonstrate proficiency in applying scientific method to models of machine learning.
- Demonstrate an ability to share in discussions of AI, its current scope and limitations, and societal implications

Unit I

Introduction to AI, Application areas of AI, State – Space Search: Production system design, Production system characteristic. Search Techniques Blind search: Depth first search, Breadth first search. Heuristic search Techniques: Hill Climbing, best first search, Branch and bound,

A*, AO* algorithms.

Unit II

Game playing: Min-Max Search, Alpha-Beta Cutoff. Knowledge Logic: Skolemizing queries, Unification algorithm, Modes Ponens, Resolution. Structured knowledge representation: Semantic nets, Frames, Conceptual dependencies, Scripts.

Unit III

Expert System: Expert System need & Justification, Rule based architecture, Nonproduction system architecture, Case studies of expert system: MYCIN, R1. Learning: Concept of learning, Types of learning, Genetic algorithm, neural network.

Unit IV

Natural language processing: Introduction syntactic processing, semantic analysis, discourse, and pragmatic processing. Handling uncertainty: Probabilistic reasoning, Use of certainty factor, Knowledge organization & management: Introduction, HAM.

Books:

Reference Books:

1. Rich & Knight (Chapters 1.1, 2, 3, 5, 9, 10.1, 10.2, 12.1, 12.2, 12.3, 15.1-15.4, 17.1-17.6)
2. Dan W. Patterson (Chapters 4.1-4.4, 5.3, 11.1, 11.4, 15.1-15.4)
3. N.J. Nilson Norvig "Artificial Intelligence a Modern Approach".